

Venous & Lymphatic drainage of lower limb

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Veins of the Lower Limb

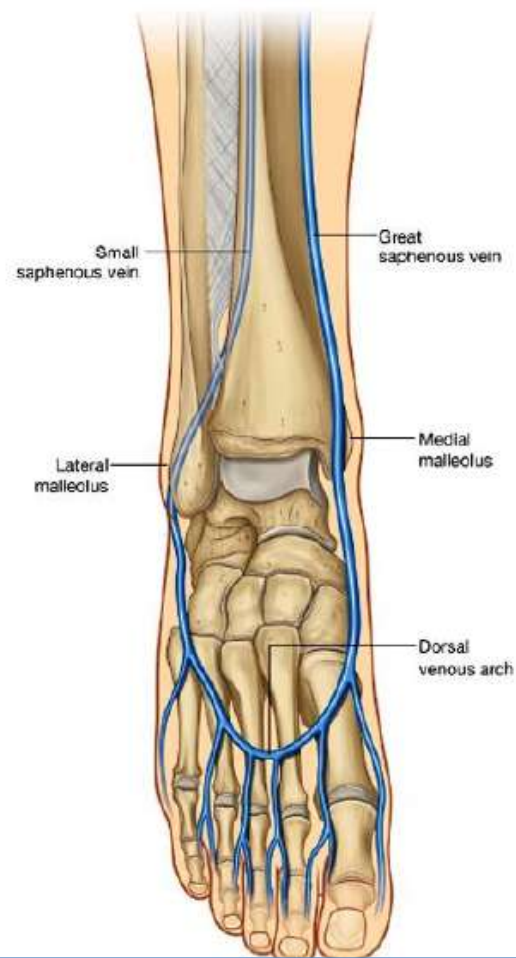
- ❖ **Divided into:** 1-superficial veins 2- deep veins 3-perforating veins.

Superficial veins

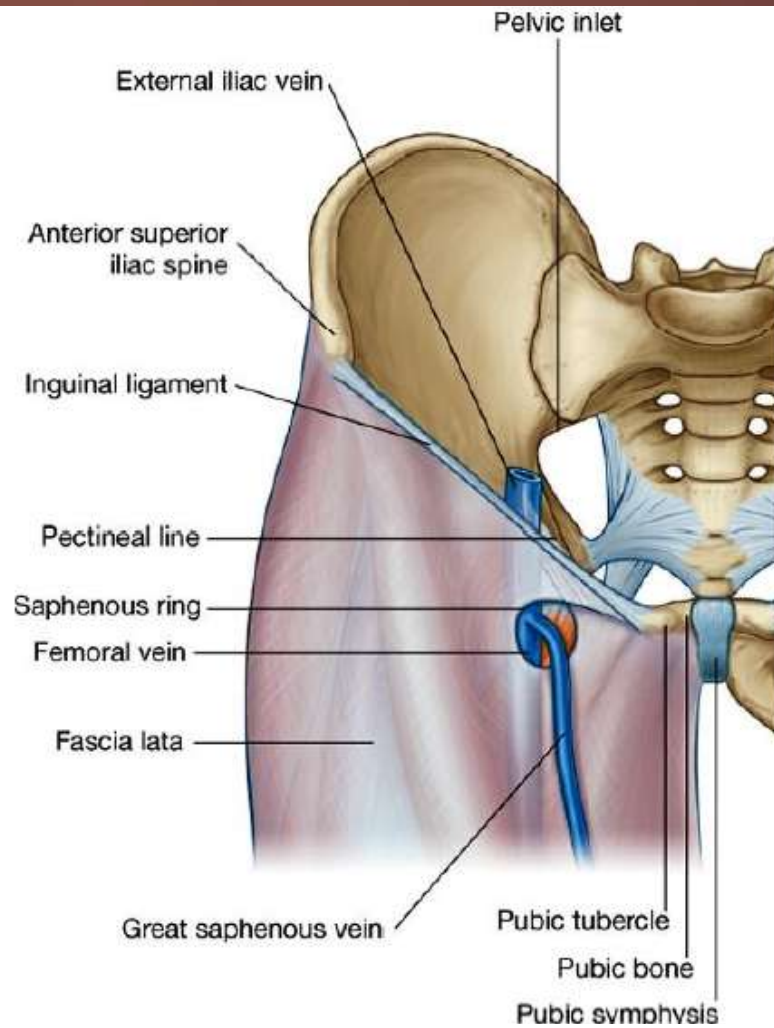
- ❖ They are in the **subcutaneous connective** tissue
- ❖ They are interconnected & ultimately **drain into the deep veins.**
- ❖ They form two major channels; Great saphenous & Small Saphenous vein and their tributaries. Both veins originate from a dorsal venous arch in the foot.
- ❖ They are of great clinical importance.

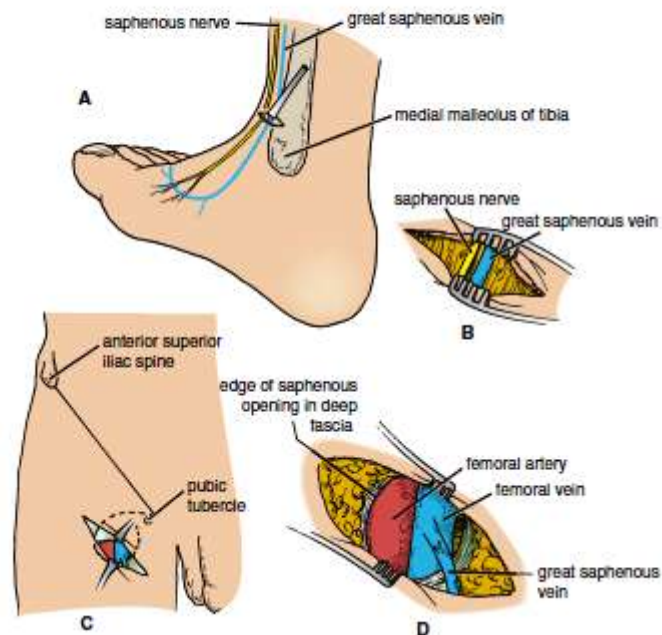
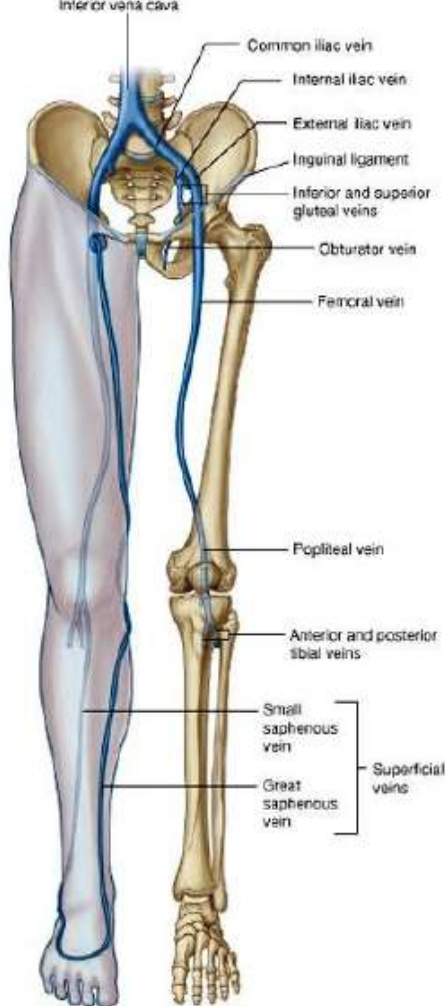
Great Saphenous vein

- **Origin:** it begins from **medial side of dorsal venous arch** on the dorsum of foot.
- **Course:**
- It passes upward directly **infront of medial malleolus.**
- It then ascends in medial side **of the leg** accompanying **Saphenous nerve** .
- It then passes behind the knee& curves forward around medial side of the thigh.
- **End:** It passes through **Saphenous Opening** in deep fascia to join **Femoral vein.**



Dorsal venous arch





○ Tributaries:

- a) superficial external pudendal vein
- b) superficial epigastric vein
- c) superficial circumflex iliac vein

Superficial inguinal veins
drained into saphenous
vein, before its
termination.

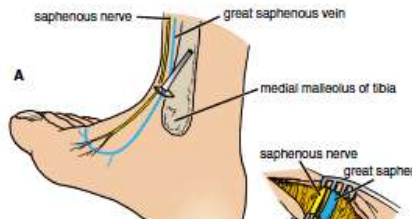
d) it is connected to **small saphenous vein** by small vein across the popliteal fossa.

e) It is connected to **deep veins** by **perforating veins** along the medial side of the calf.

Great saphenous vein cutdown



The constant position of the great saphenous vein in front of the medial malleolus, should be remembered for patients requiring emergency blood transfusion.



Small Saphenous vein

1-Origin: it begins from **lateral side of dorsal venous arch** on the dorsum of foot.

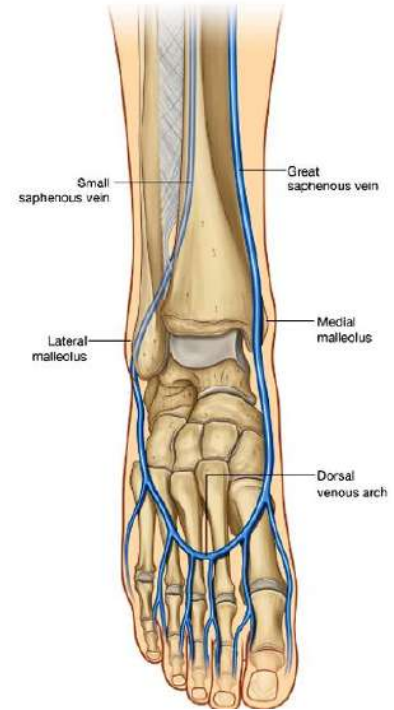
2-Course: It ascends **behind the lateral malleolus**, in company with **Sural nerve**.

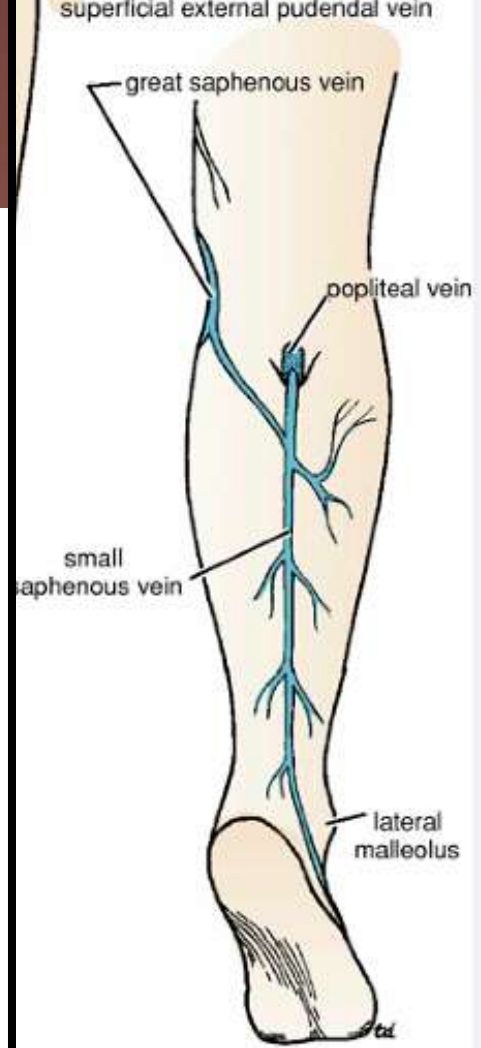
- Then runs up the middle of the back of the leg.
- It then pierces deep fascia between 2 heads of gastrocnemius.

3- End: in the popliteal fossa, it ends in **popliteal vein**.

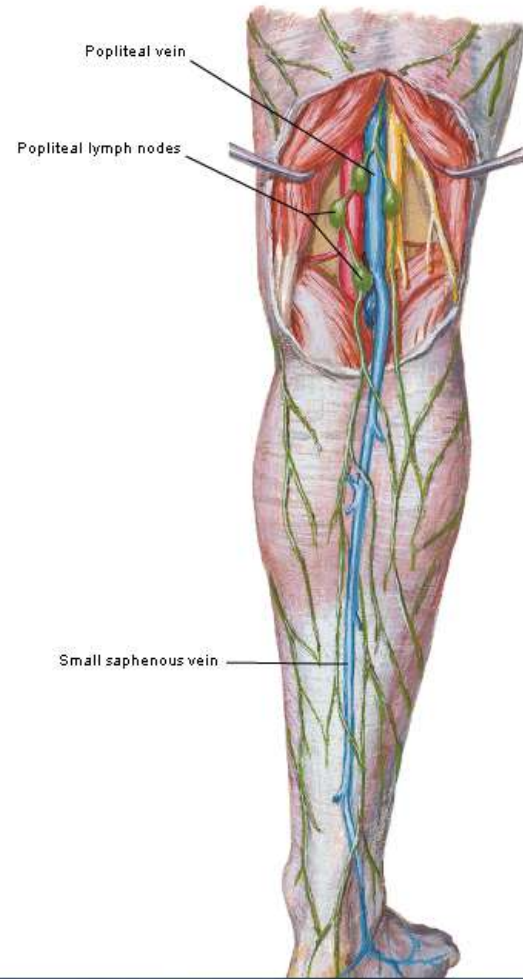
4-Tributaries:

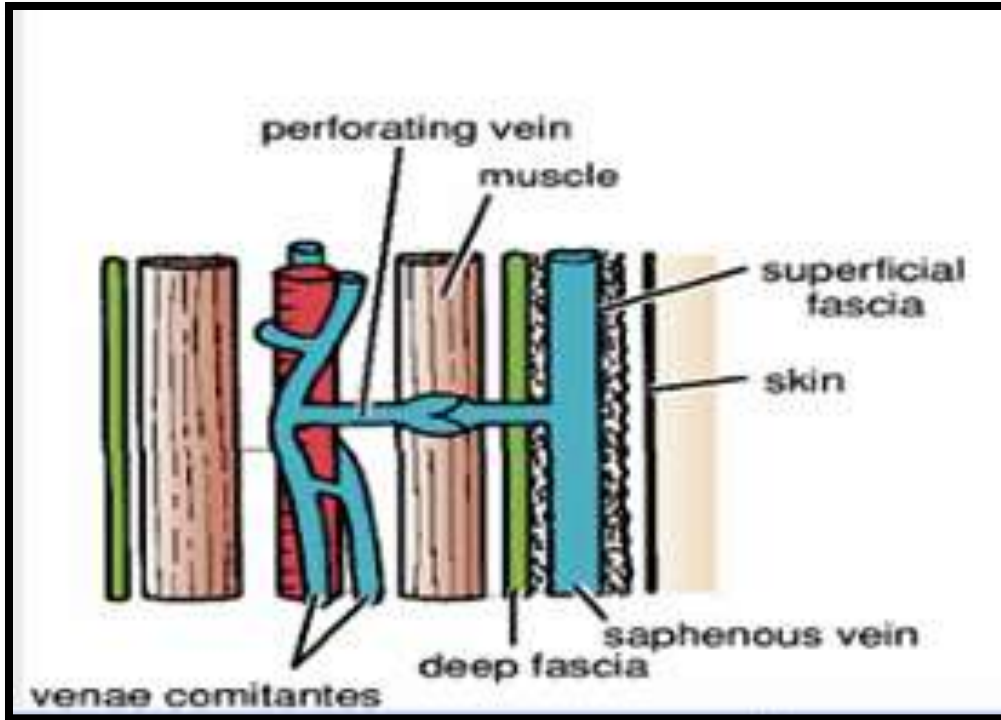
- numerous small veins from back of the leg.
- Along its course, it connected with deep veins in the leg & foot by perforating veins
- It is connected with great saphenous vein by small vein across popliteal fossa.





Lymph Vessels and Nodes of Lower Limb Posterior View





The normal flow of blood in the lower limb, is from the skin & subcutaneous tissues to the **superficial veins**, which drain via **perforating veins (contain one way valves)** to the **deep veins**, which in turn drain into **iliac veins & IVC**.

- Venous return is supplemented with **muscle contraction**, which pumps the blood towards the heart
- The normal flow of blood in the venous system depends on the **presence of competent valves**, which prevent Reflux الإرتجاع

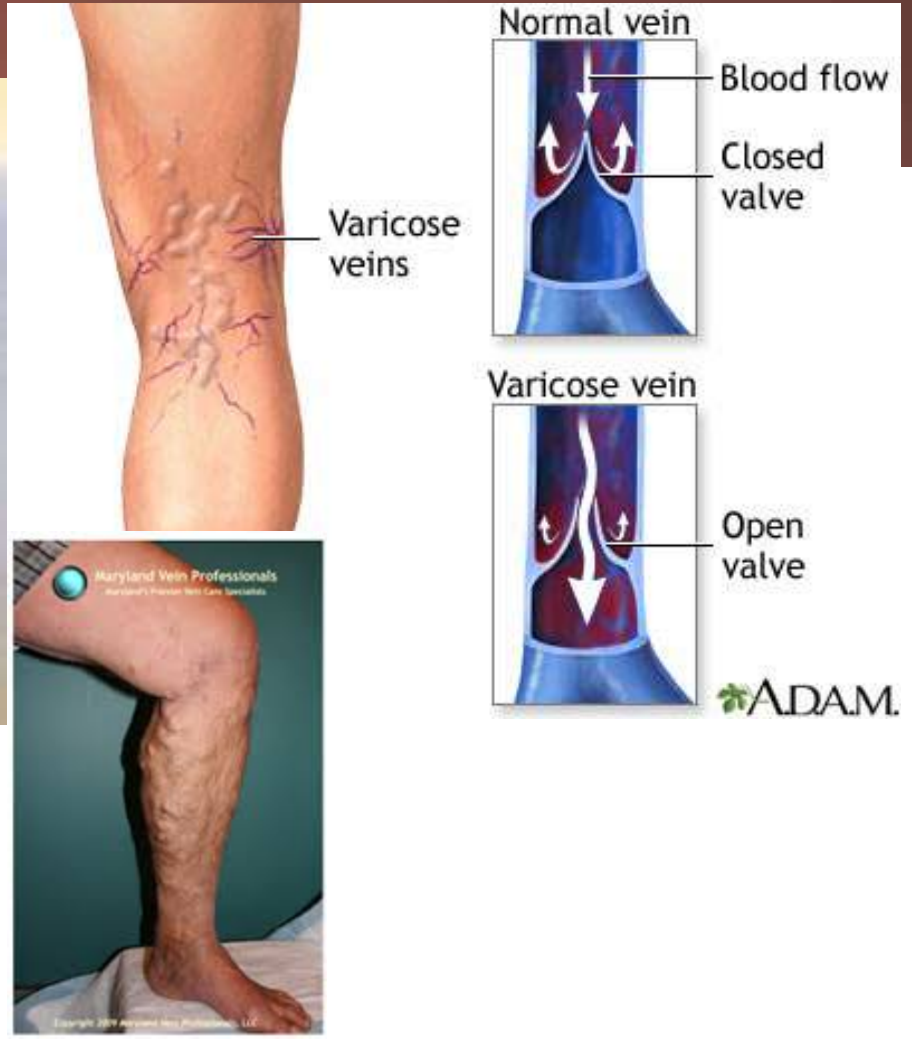
Varicose veins الدوالي



- When the venous valves become incompetent, they tend to place extra pressure on more distal valves, which may also become incompetent....this will produce dilated tortuous superficial veins (varicose veins) in the distribution of short and long saphenous venous system.
- More common in women than men .
- Treatment:** stripping long&short saphenous veins or valvular reconstruction.



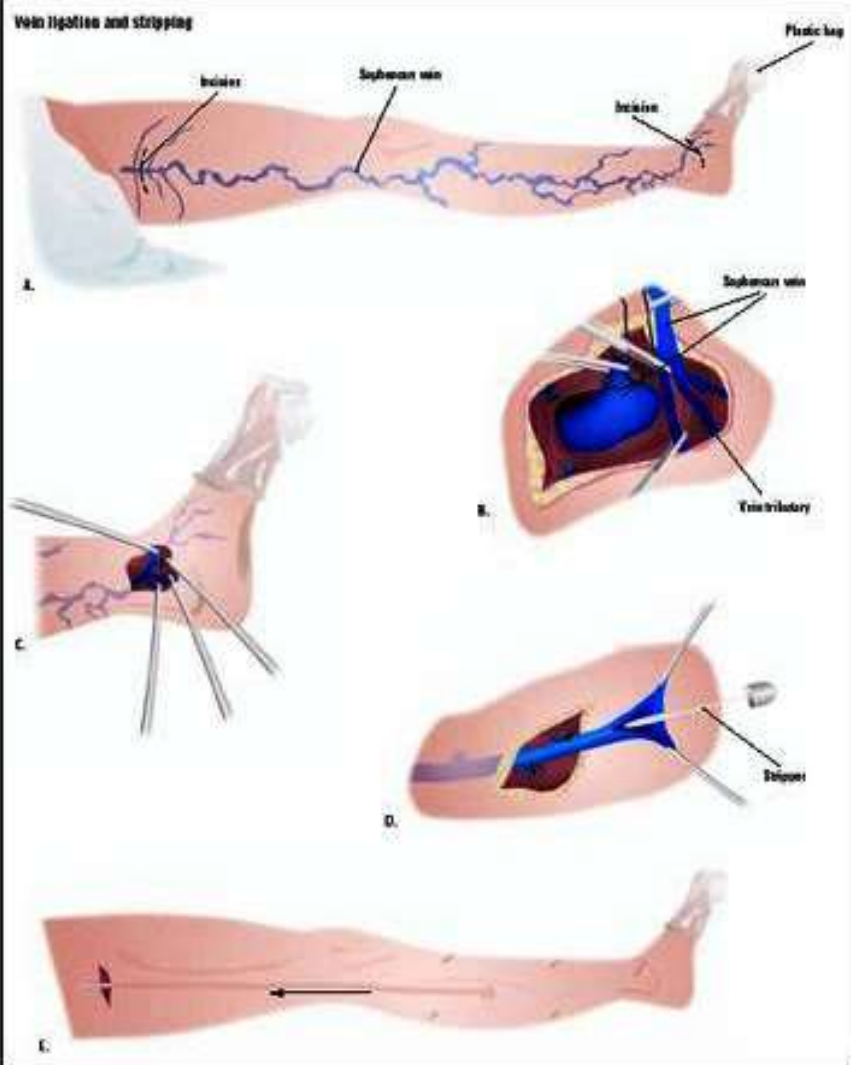
Varicose veins





Deep vein thrombosis

Vein ligation and stripping

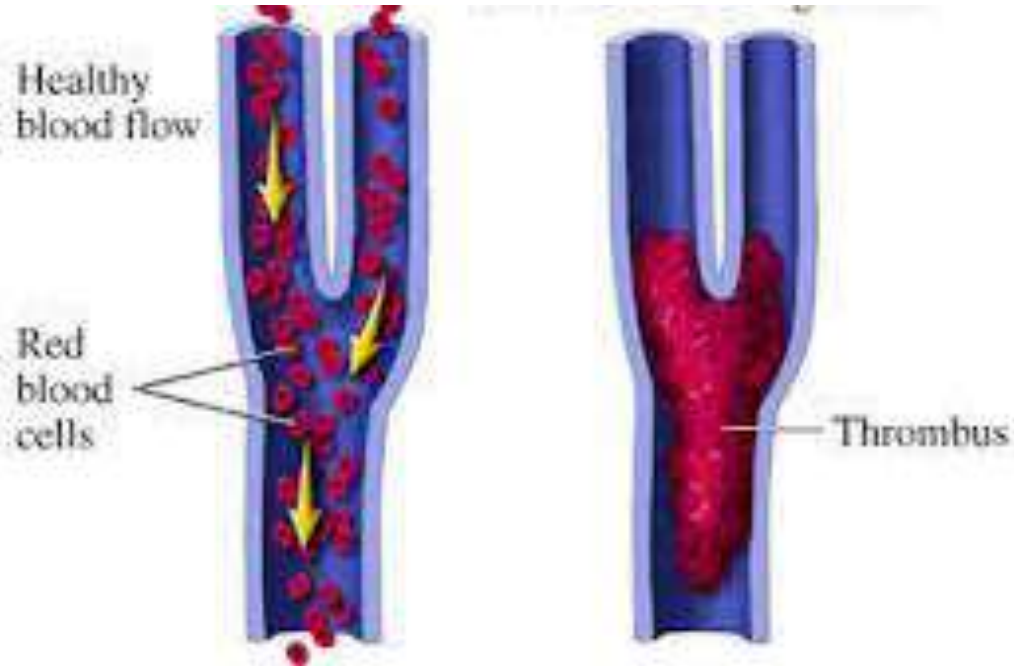


جأطة وريدیه (DVT) Deep vein Thrombosis

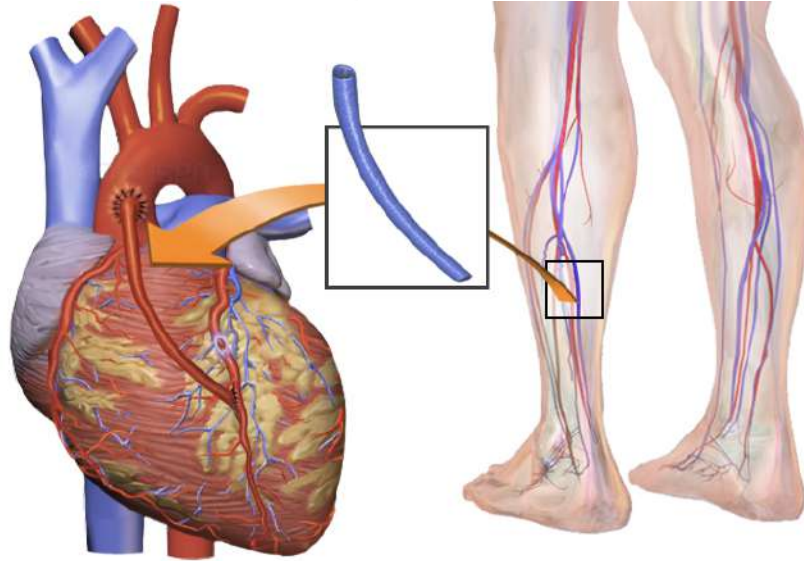


- Sitting immobile for hours(long time air travel), may cause DVT in the leg.
- Thrombosis in the veins of soleus muscle gives mild pain or tightness in the calf & calf muscle tenderness.
- It can occur with no signs or symptoms.
- If thrombosis is dislodged, it passes to heart & lungs, which is **often fatal**.
- Stretching leg every hour to improve venous return is important.

Deep vein thrombosis



Coronary By pass surgery



In patients with occlusive coronary heart disease, the diseased arterial segment can be by passed by inserting graft consisting of a portion of great saphenous vein.

❖ Deep veins of the lower limb

- ❖ Are the vena comitantes to the anterior & posterior tibial arteries, the popliteal vein, and the femoral vein and their tributaries.

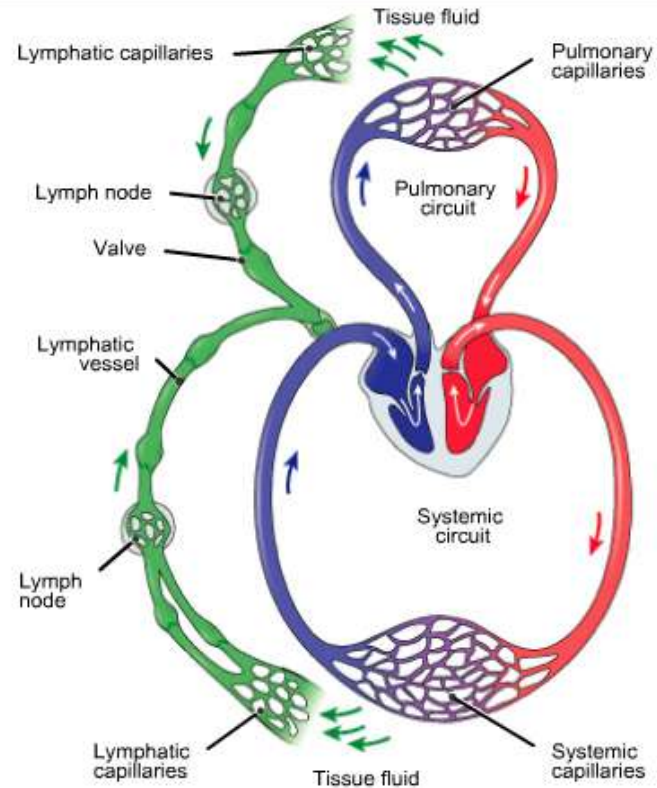
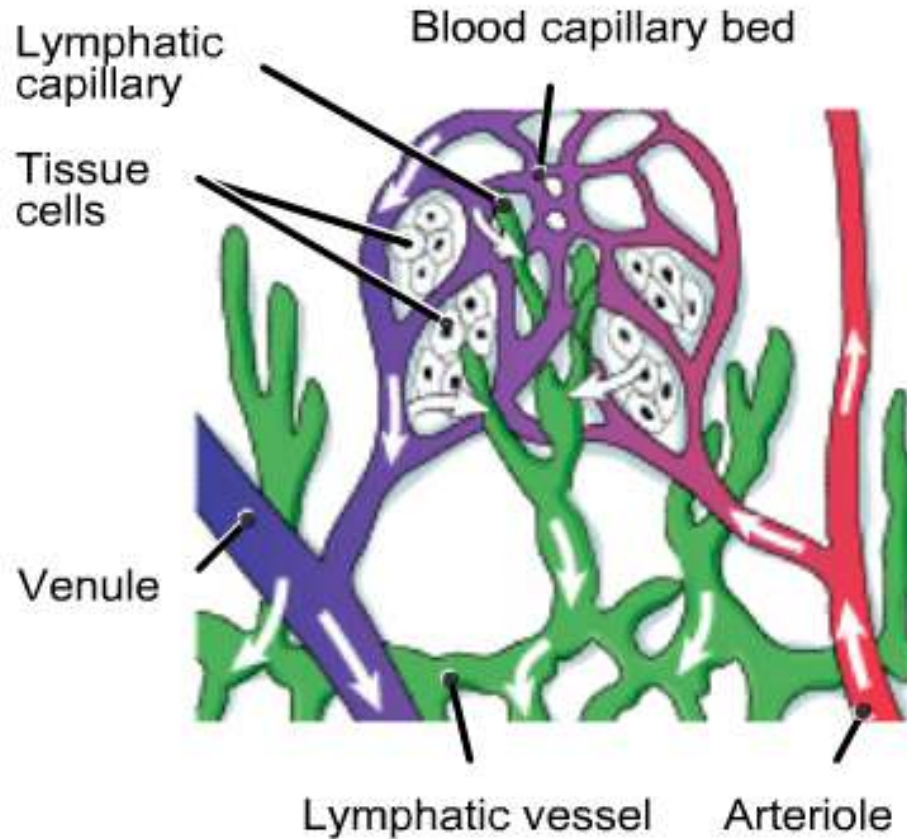
❖ Perforating veins:

- Are communicating vessels that run between superficial and deep veins.
- They possess **valves** that are arranged to prevent the flow of blood from the deep to superficial veins.



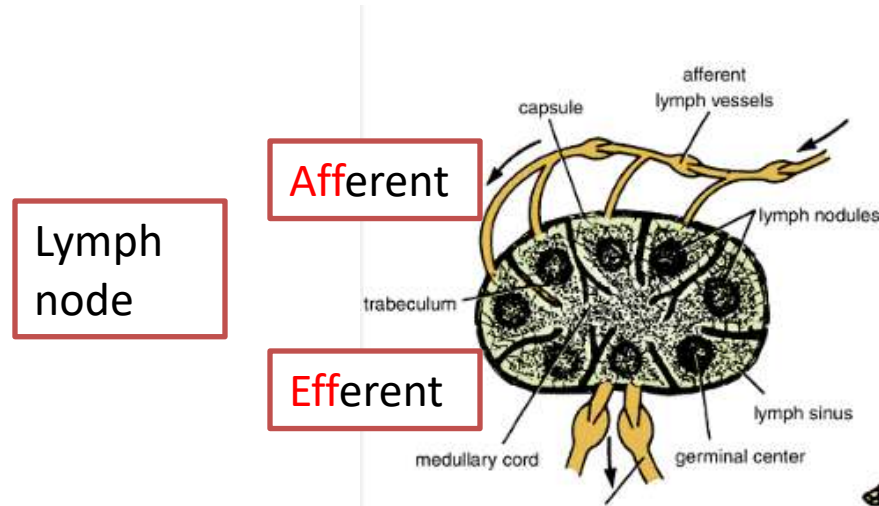
Lymphatic drainage of lower limb.

- ❖ السائل اللمفاوي? Lymph
- ❖ عقد لمفاوية? Lymph node
- ❖ أوعية لمفاوية? Lymph vessels or lymphatics



Lymph: (L. lymph = clear water)

- Lymph is a clear colourless fluid which is formed at the tissue spaces (intercellular spaces).
- It is collected by lymph capillaries from the intercellular spaces



Inguinal lymph nodes

A-Superficial inguinal lymph nodes

- Lie in the superficial fascia below the inguinal ligament & can be divided into a 1) horizontal group 2) vertical group
- The efferent lymph vessels from the **superficial inguinal nodes** pass through the saphenous opening in the deep fascia and join the deep inguinal nodes.

Superficial inguinal lymph nodes

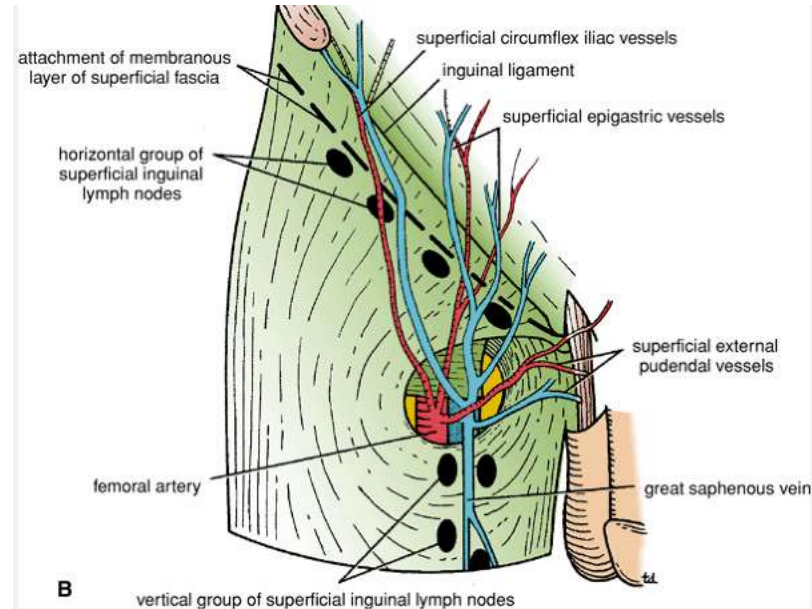
A-Horizontal group: lies just below & parallel to inguinal ligament

1-Medial members:

receive superficial lymph vessels from the **anterior abdominal wall below the level of the umbilicus, perineum, urethra, external genitalia of both sexes (but not the testes) and lower 1/2 of anal canal.**

2-Lateral members:

receive superficial lymph vessels **from the back below the level of the iliac crests.**

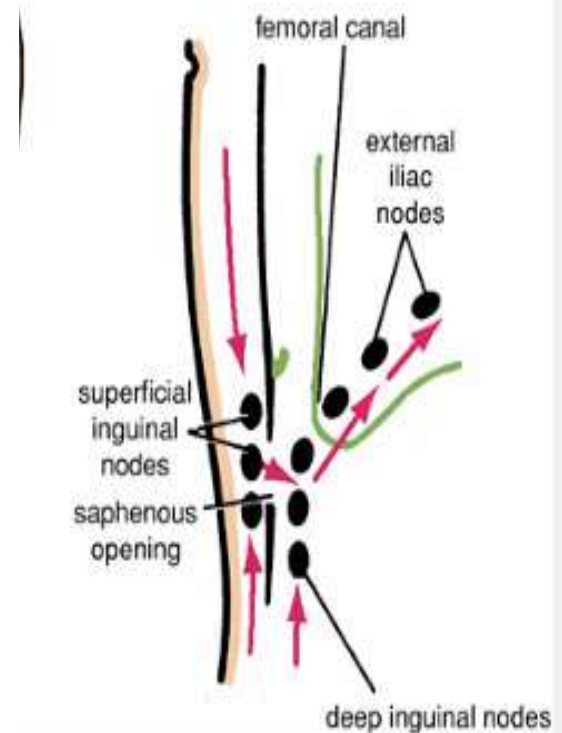


B- Vertical group

lies along the **terminal part of the great saphenous vein**, and receives most of the superficial lymph vessels of the lower limb.

B-Deep inguinal lymph nodes

- are located beneath the deep fascia and lie along **the medial side of the femoral vein**; the efferent vessels from these nodes enter the abdomen by passing through the femoral canal to **lymph nodes** along the **external iliac artery**.



- ❑ **The superficial inguinal LNs** not only drain all the lymph from the lower limb, but also drain lymph from:
 - ❑ the skin and superficial fascia of the anterior and posterior abdominal walls below the level of the umbilicus & below level of iliac crest respectively.
 - ❑ lymph from the external genitalia and the mucous membrane of the lower half of the anal canal.
 - ❑ Back, below level of iliac crest.

